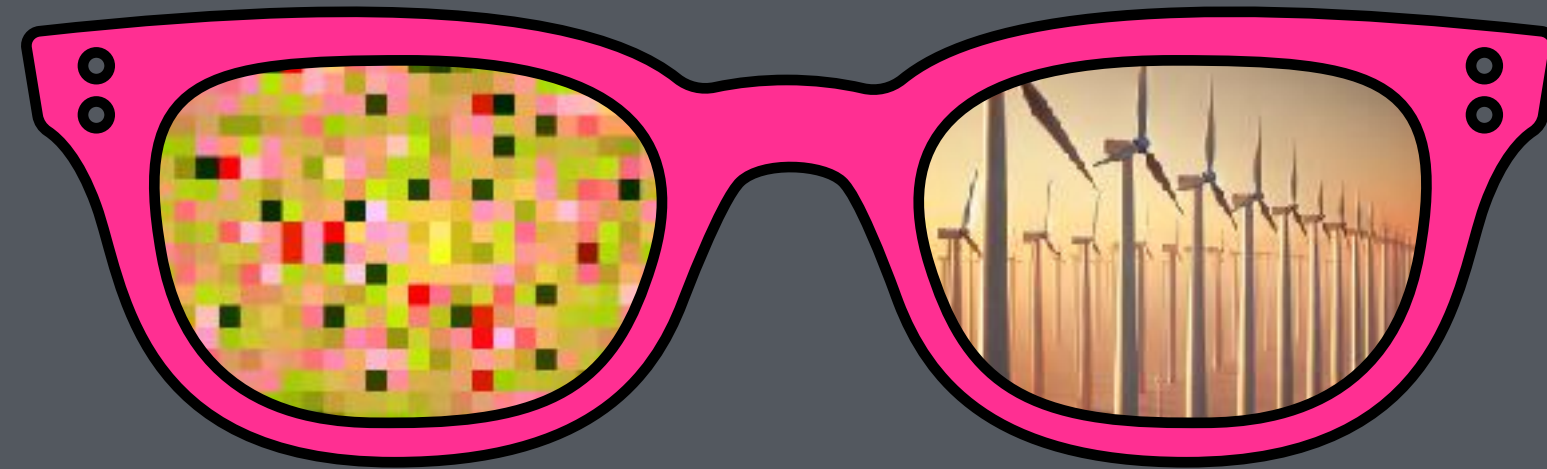


Data and the State

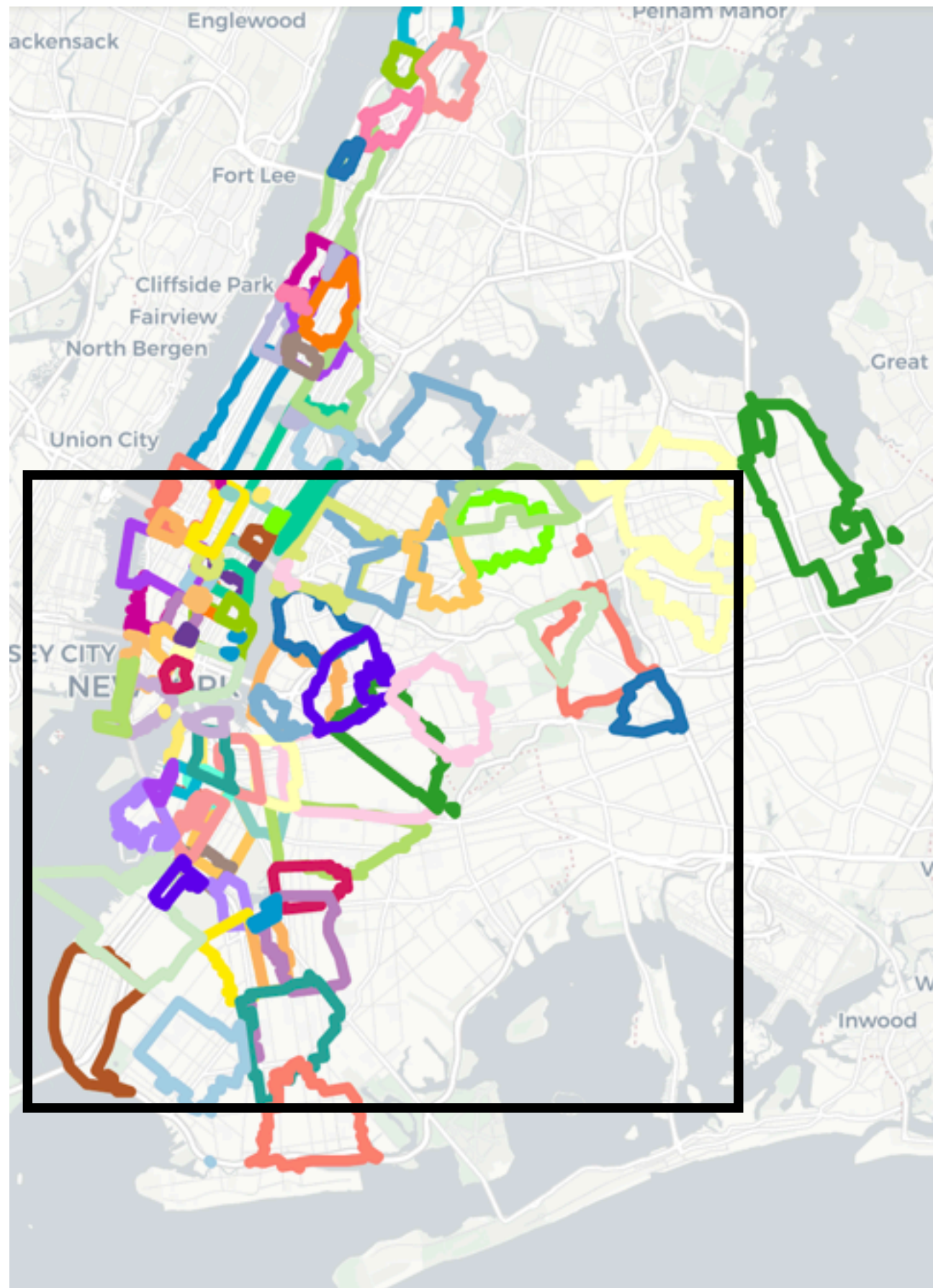
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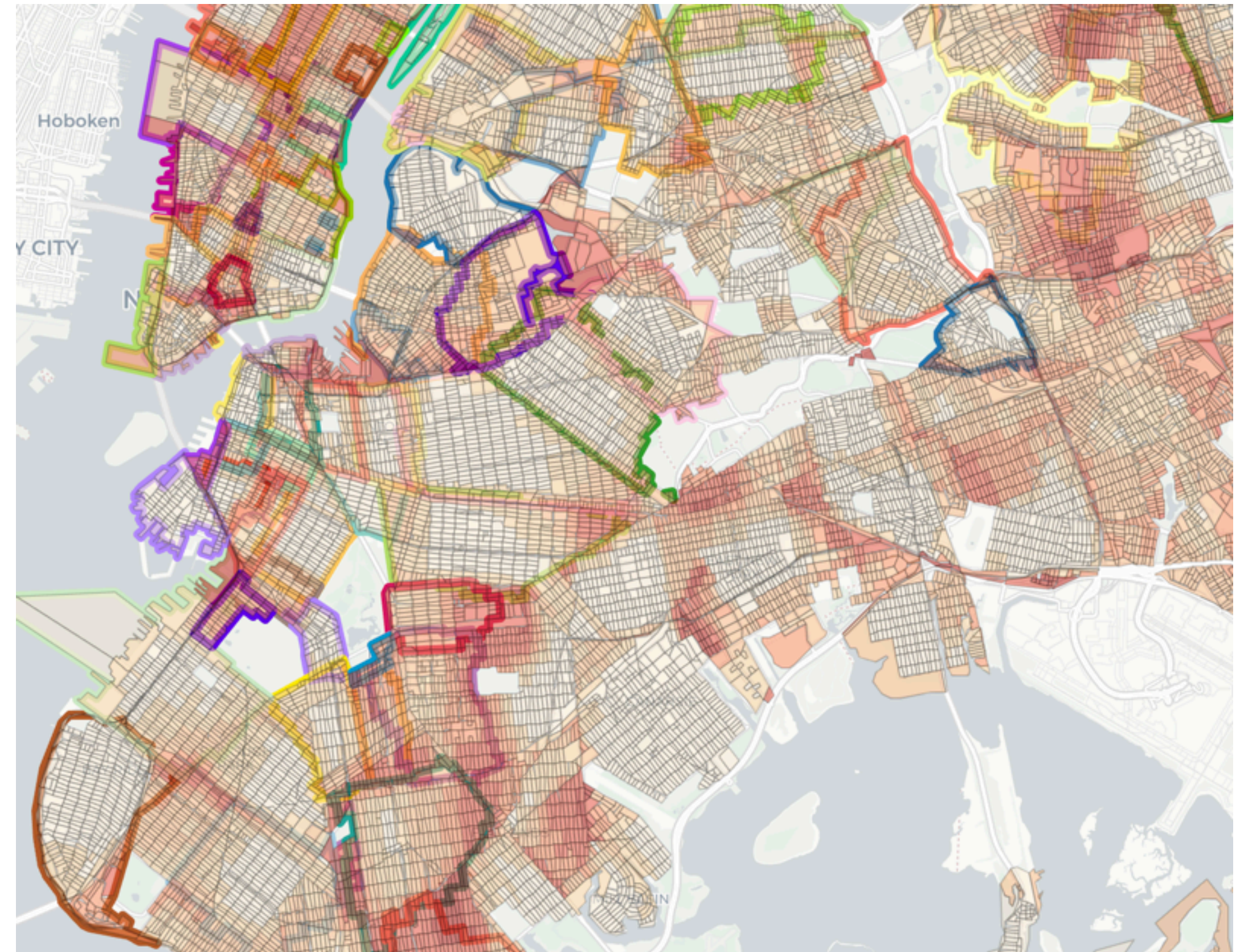
Thick and thin:
Communities and state power

Lecture 18, Wednesday May 20

cores



cores overlaid with entropy heatmap



(how uncertain is the label?)

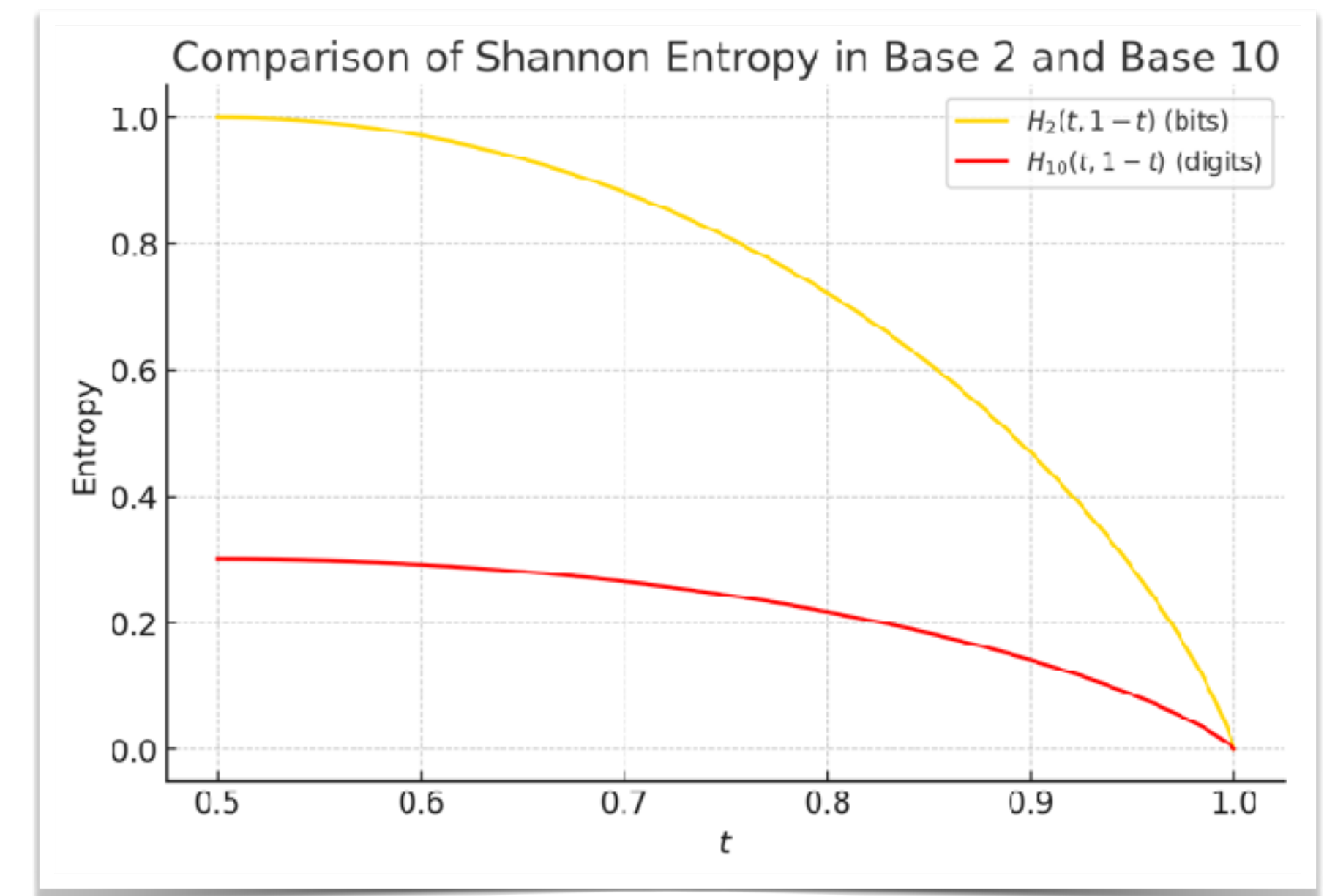
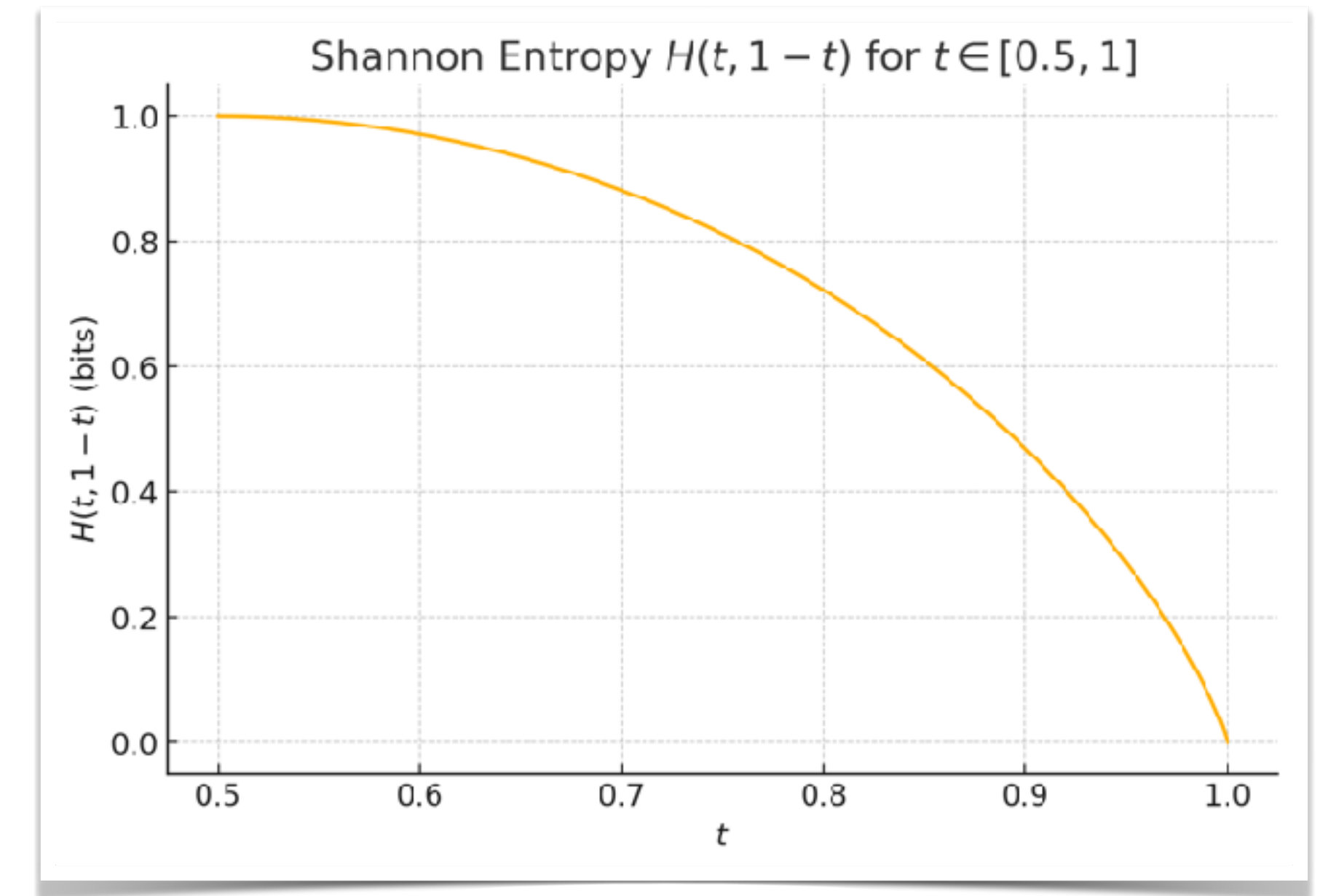
How entropy works

$$H(w) = - \sum w_i \log w_i$$

- Minimum number of bits or digits needed to encode something $\leftarrow$ *those are the units!!*



Figure 1: It takes a lot of bits to give a precise description of a chaotic configuration.



By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have meaning; that is, they are correlated according to some system with certain physical states of affairs. The aspects of communication are irrelevant to the engineering problem. The message is one selected from a set of possible messages. The problem is to reproduce the message, not just the one which will actually be chosen.

If the number of messages in the set is finite then this number can be regarded as a measure of the information produced with the choices being equally likely. As was pointed out by Hartley, this is a logarithmic function. Although this definition must be generalized considerably to include the statistics of the message and when we have a continuous range of possibilities, it is essentially logarithmic measure.

The logarithmic measure is more convenient for various reasons:

1. It is practically more useful. Parameters of engineering importance such as time, bandwidth of relays, etc., tend to vary linearly with the logarithm of the number of possibilities. For example, adding one relay to a group doubles the number of possible states of the relays. It adds 1 to the logarithm of this number. Doubling the time roughly squares the number of possible messages, which doubles the logarithm, etc.
2. It is nearer to our intuitive feeling as to the proper measure. This is closely related to (1). Intuitively measures entities by linear comparison with common standards. One feels, for example, that two punched cards should have twice the capacity of one for information storage, and two channels twice the capacity of one for transmitting information.
3. It is mathematically more suitable. Many of the limiting operations are simple in terms of logarithms but would require clumsy restatement in terms of the number of possibilities.

The choice of a logarithmic base corresponds to the choice of a unit for measuring information. If base 2 is used the resulting units may be called binary digits, or more briefly *bits*, a word suggested by J. W. Tukey. A device with two stable positions, such as a relay or a flip-flop circuit, can store one bit of information. N such devices can store N bits, since the total number of possible states is 2^N and $\log_2 2^N = N$. If the base 10 is used the units may be called decimal digits. Since

$$\begin{aligned}\log_2 M &= \log_{10} M / \log_{10} 2 \\ &= 3.32 \log_{10} M,\end{aligned}$$

continuous

monotonic

composable

If there is such a measure, say $H(p_1, p_2, \dots, p_n)$, it is reasonable to require of it the following properties:

1. H should be continuous in the p_i .
2. If all the p_i are equal, $p_i = \frac{1}{n}$, then H should be a monotonic increasing function of n . With equally likely events there is more choice, or uncertainty, when there are more possible events.
3. If a choice be broken down into two successive choices, the original H should be the weighted sum of the individual values of H . The meaning of this is illustrated in Fig. 6. At the left we have three

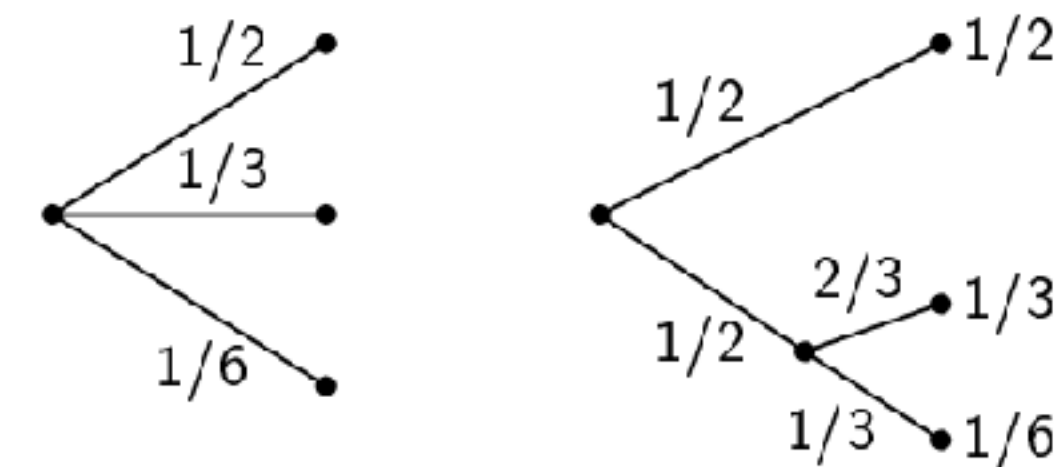
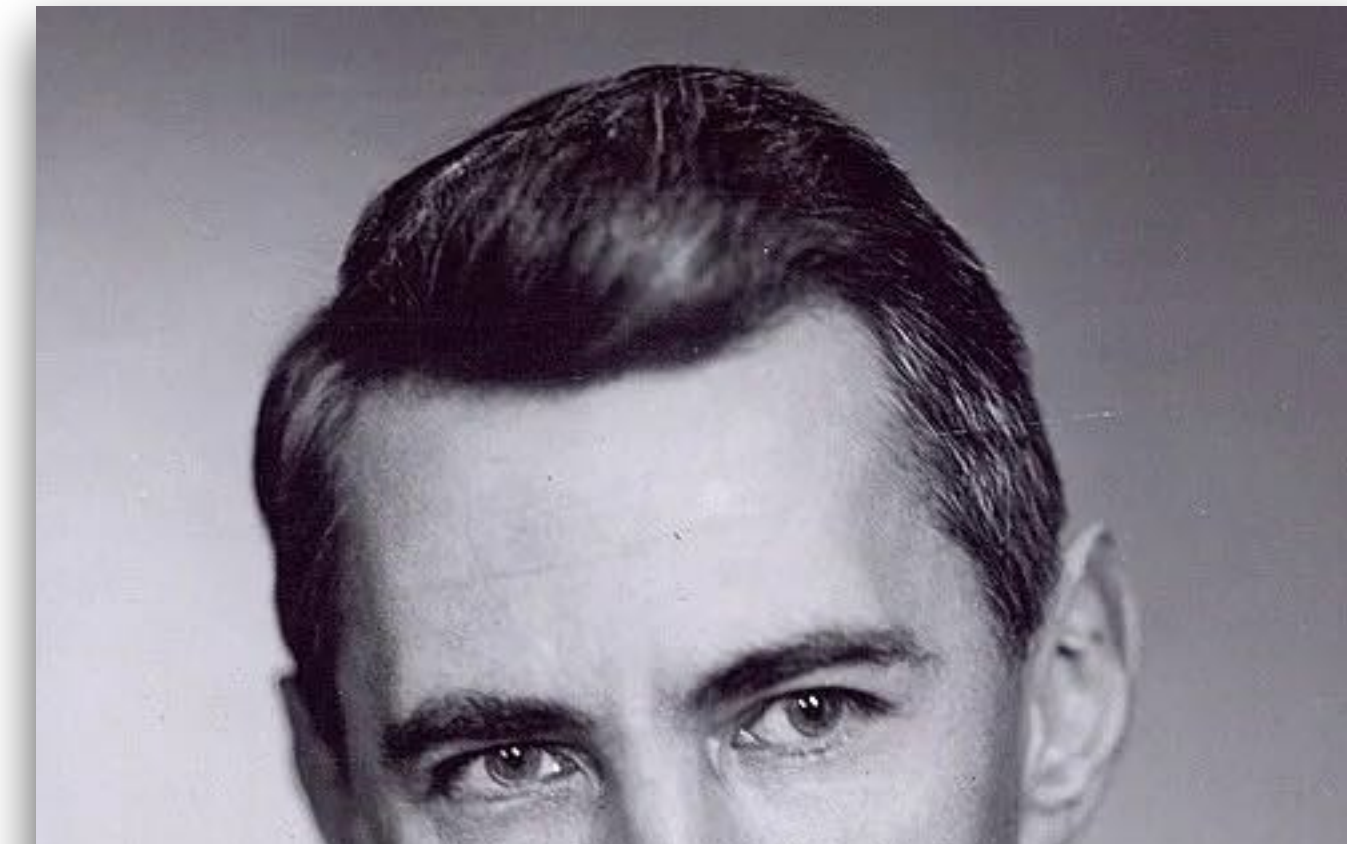


Fig. 6—Decomposition of a choice from three possibilities.

possibilities $p_1 = \frac{1}{2}$, $p_2 = \frac{1}{3}$, $p_3 = \frac{1}{6}$. On the right we first choose between two possibilities each with probability $\frac{1}{2}$, and if the second occurs make another choice with probabilities $\frac{2}{3}$, $\frac{1}{3}$. The final results have the same probabilities as before. We require, in this special case, that

$$H\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) = H\left(\frac{1}{2}, \frac{1}{2}\right) + \frac{1}{2}H\left(\frac{2}{3}, \frac{1}{3}\right).$$



Community

- **THICK:** “I have called my tiny community a world, and so its isolation made it; and yet there was among us but a half-awakened common consciousness, sprung from common joy and grief, at burial, birth, or wedding; from a common hardship in poverty, poor land, and low wages; and, above all, from the sight of the Veil that hung between us and Opportunity.” – W.E.B. Du Bois, *The Souls of Black Folk* .

- **THIN:** Shared measurable attributes, like income, transit use, etc
- Thick vs thin description comes from anthropology (e.g., Clifford Geertz on **culture** in 1970s) and permeates social science

shared needs and worries



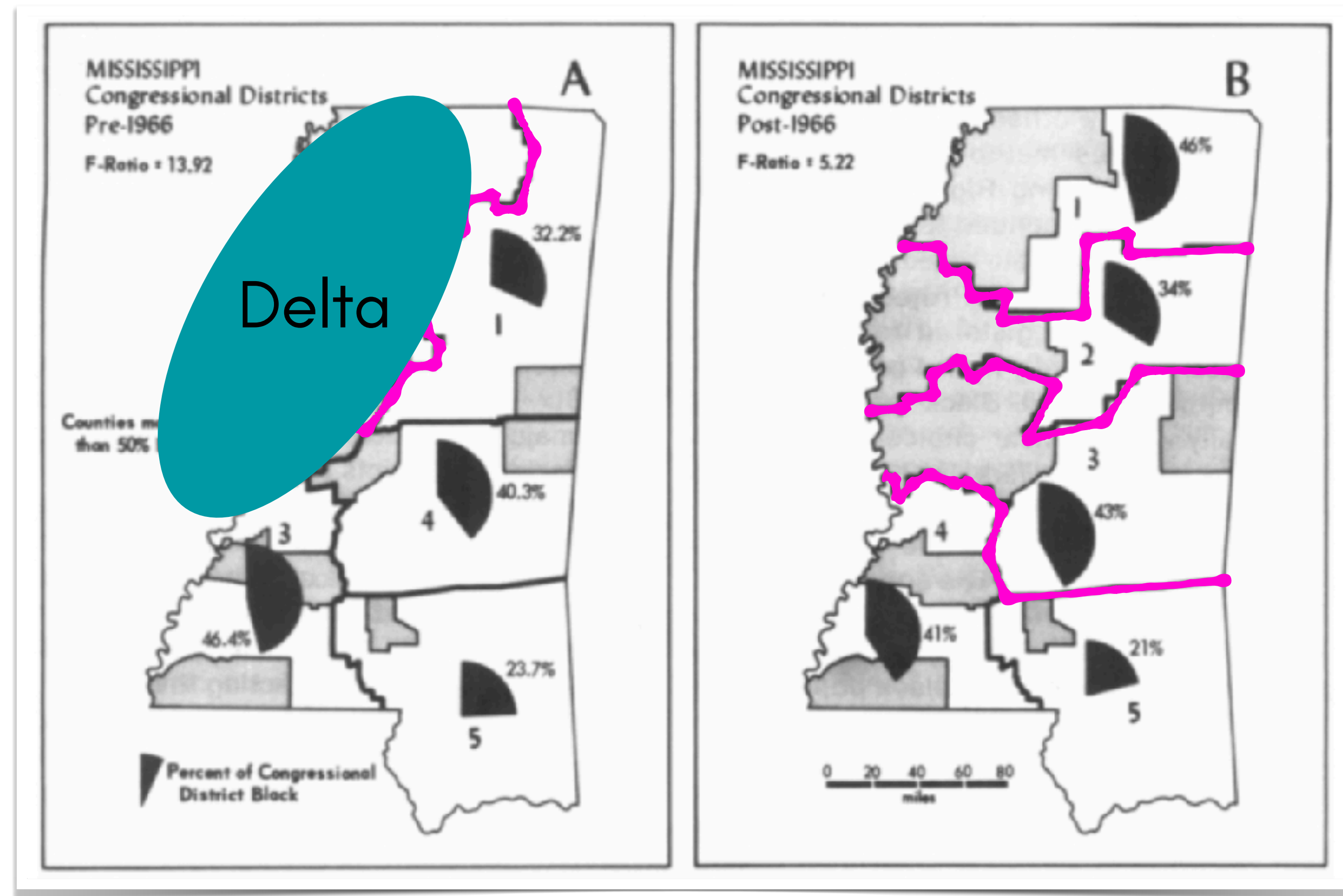
shared interests/industries



layering state power on top of
community formations

What is redistricting?

- Dividing up a locality for an administrative purpose: services, schools, **elections**
- There are 90,000 local governments in the U.S.!
- But when you know a lot about preferences of voters, this is obviously manipulable through **gerrymandering**



“Communities of interest”

- **Idaho:** “To the maximum extent possible, districts shall preserve traditional neighborhoods and local communities of interest.”
- **Oregon:** “Each district, as nearly as practicable, shall ... Not divide communities of common interest.”
- **South Dakota:** “Protection of communities of interest by means of compact and contiguous districts.”

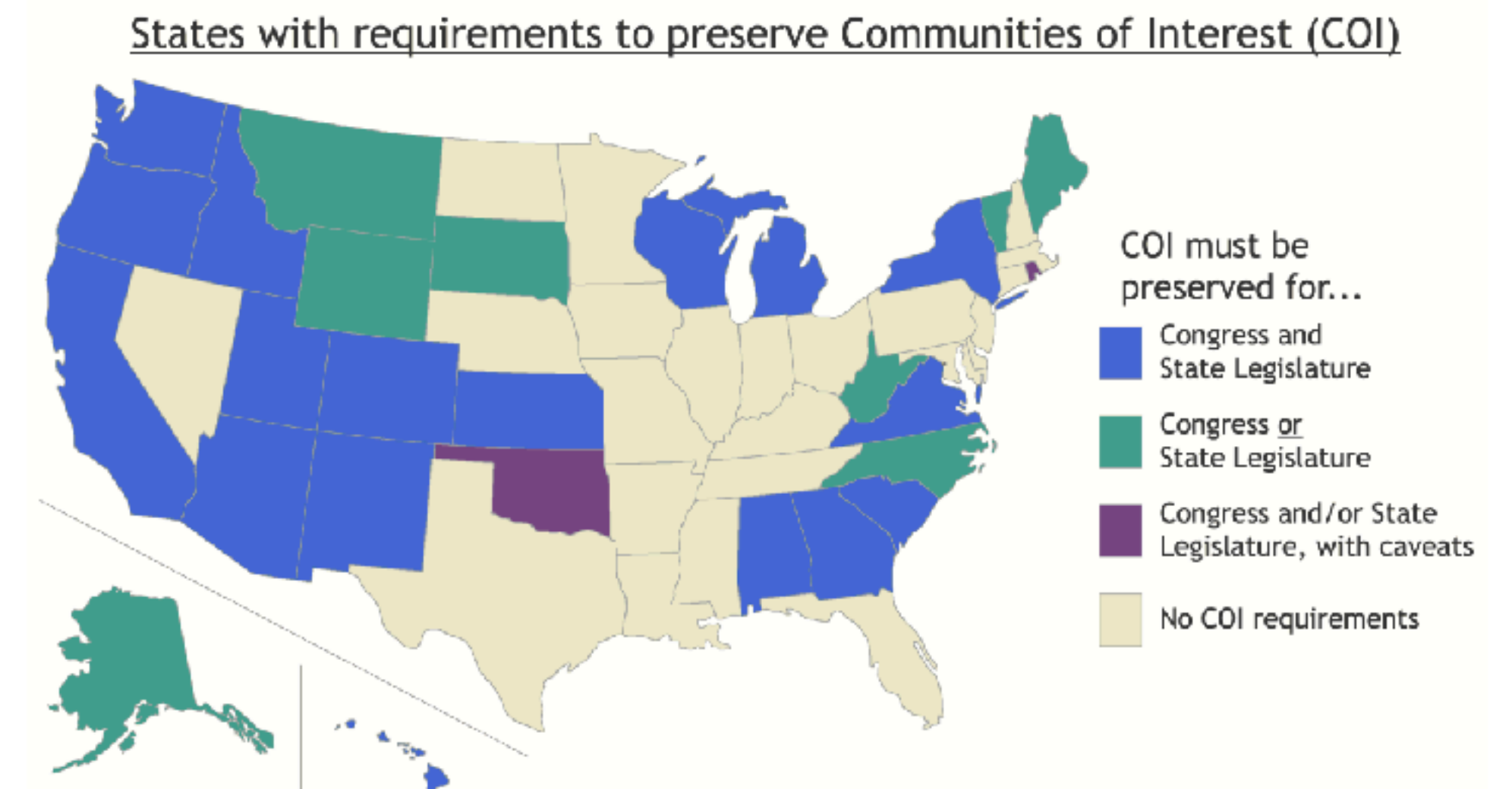


Figure 1: Community of Interest language in the rules for redistricting. Data from [1, 10].

“Communities of interest”

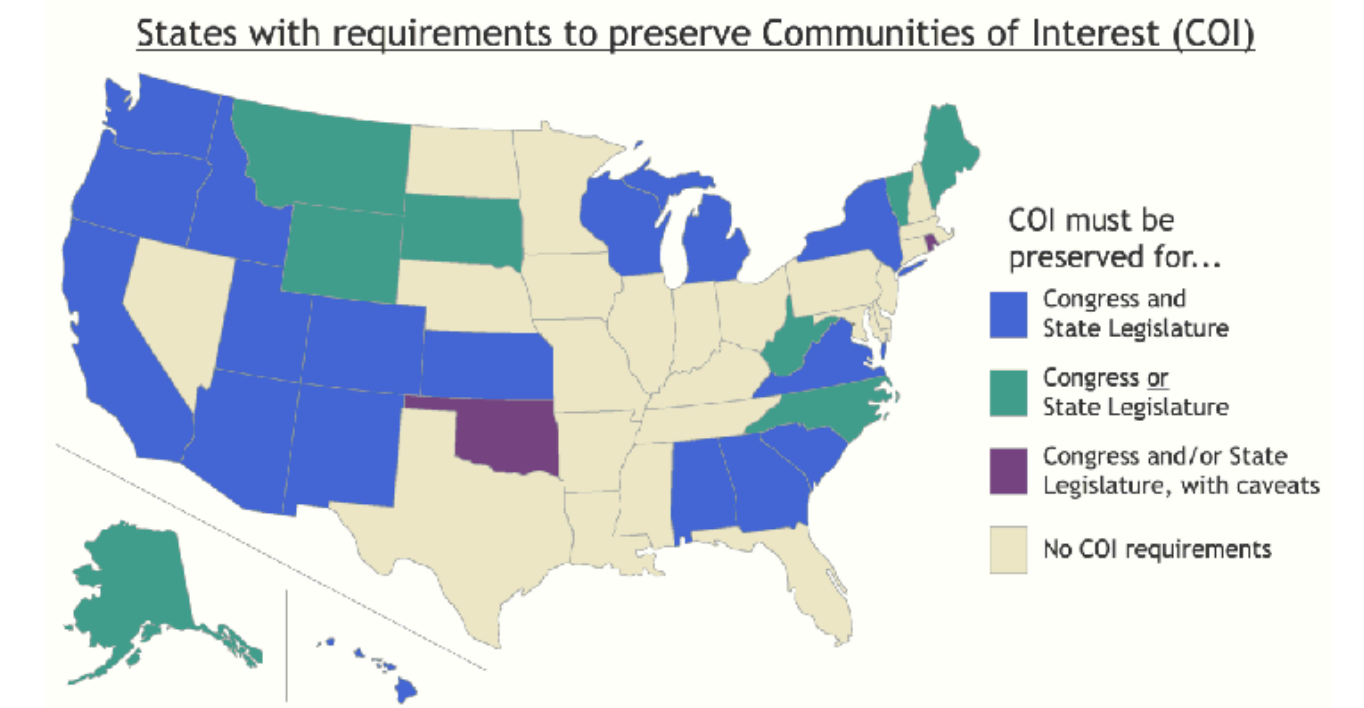


Figure 1: Community of Interest language in the rules for redistricting. Data from [1, 10].

- **Alaska:** “...a relatively integrated socio-economic area.”
- **California:** “A community of interest is a contiguous population which shares common social and economic interests that should be included within a single district for purposes of its effective and fair representation. Examples of such shared interests are those common to an urban area, a rural area, an industrial area, or an agricultural area, and those common to areas in which the people share similar living standards, use the same transportation facilities, have similar work opportunities, or have access to the same media of communication relevant to the election process. Communities of interest shall not include relationships with political parties, incumbents, or political candidates.”
- **Colorado:** “communities of interest, including ethnic, cultural, economic, trade area, geographic, and demographic factors, shall be preserved within a single district wherever possible.”
- **Alabama:** “community of interest is defined as an area with recognized similarities of interests, including but not limited to racial, ethnic, geographic, governmental, regional, social, cultural, partisan, or historic interests; county, municipal, or voting precinct boundaries; and commonality of communications.”

The following descriptions have counted as COIs:

- Justice Sandra Day O'Connor, writing for the Supreme Court plurality, cites "for example, shared broadcast and print media, public transport infrastructure, and institutions such as schools and churches" as arguable "manifestations of community of interest," but warns that these are not easily disentangled from race (*Bush v. Vera*, 517 U.S. at 964 (1996))
- A "predominantly urban, low-income population" (*Lawyer v. Dept of Justice*, 521 U.S. 567, 581–82 (1997));
- Satisfactory evidence of shared socio-economic status (*Chen v. City of Houston*, 206 F.3d 502, 513 (5th Cir. 2000));
- "[L]ess-educated" citizens, or those "more often unemployed" have "common social and economic needs" suitable for a COI (*Theriot v. Parish of Jefferson*, 185 F.3d 477, 486 (5th Cir. 1999));
- "There are no doubt religious, class, and social communities of interest that cross county lines and whose protection might be a legitimate consideration in districting decisions." (*Kelley v. Bennett*, 96 F. Supp. 2d 1301 (M.D. Ala.))
- Lower courts have affirmed that Latino or Hispanic groups can constitute a COI (*Barnett v. City of Chicago*, 141 F.3d 699, 704 (7th Cir. 1998) and *Meza v. Galvin*, 322 F. Supp. 2d 52, 75 (D. Mass. 2004)), but there are ample indications that the Supreme Court would not agree (*Miller*, *Session*, etc.).
- Census tracts can serve as evidence: the California Supreme Court approved the plan of an outside expert (or "special master"), in part on the reasoning that building it out of whole census tracts contributed to preserving COIs (*Wilson v. Eu*, 823 P.2d 545, 556 (Cal. 1992)).

On the other hand, the following have been rejected as a basis for COIs:

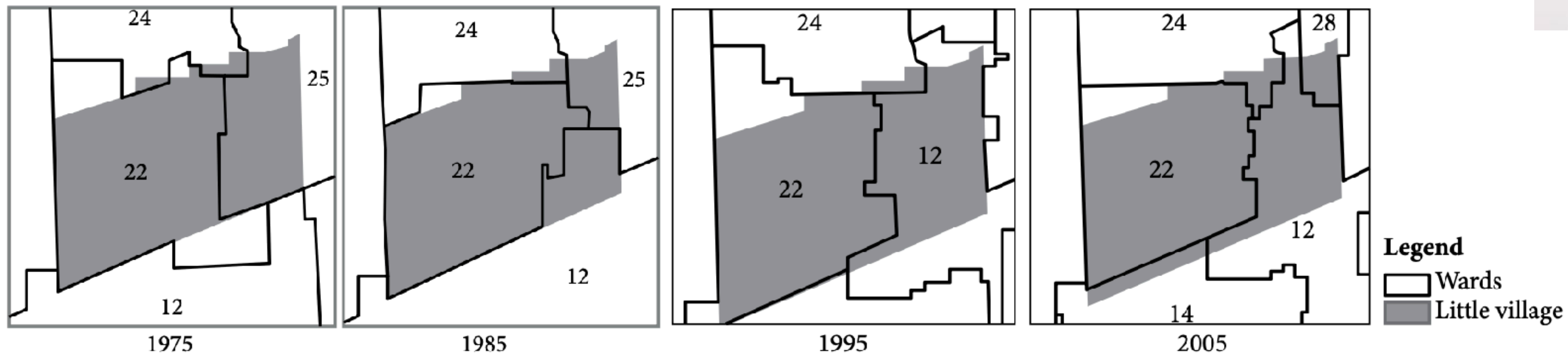
- Keeping urban apart from rural areas (*In re Legislative Districting of Gen. Assembly*, 193 N.W.2d 784, 789 (Iowa 1972));
- A court itself may not "define what a community of interest is and where its boundaries are" (*In re Legislative Districting of State*, 805 A.2d 292, 297, 298 (Md. 2002)).

Finally, heterogeneity itself—just the fact that the district is not cohesive—is sometimes taken to show that race or ethnicity predominated over COIs as criteria for drawing districts, as in the following:

- Evidence of "fractured political, social, and economic interests" argues against specific COIs (*Miller v. Johnson*, 515 U.S. 900, 919 (1995));
- "Plaintiffs presented evidence of differences in socio-economic status, education, employment, health, and other characteristics between Hispanics who live near Texas's southern border and those who reside in Central Texas" in order to argue that ethnicity had predominated over COIs in forming the districts (*Session v. Perry*, 298 F. Supp. 2d 451, 512 (E.D. Tex. 2004));

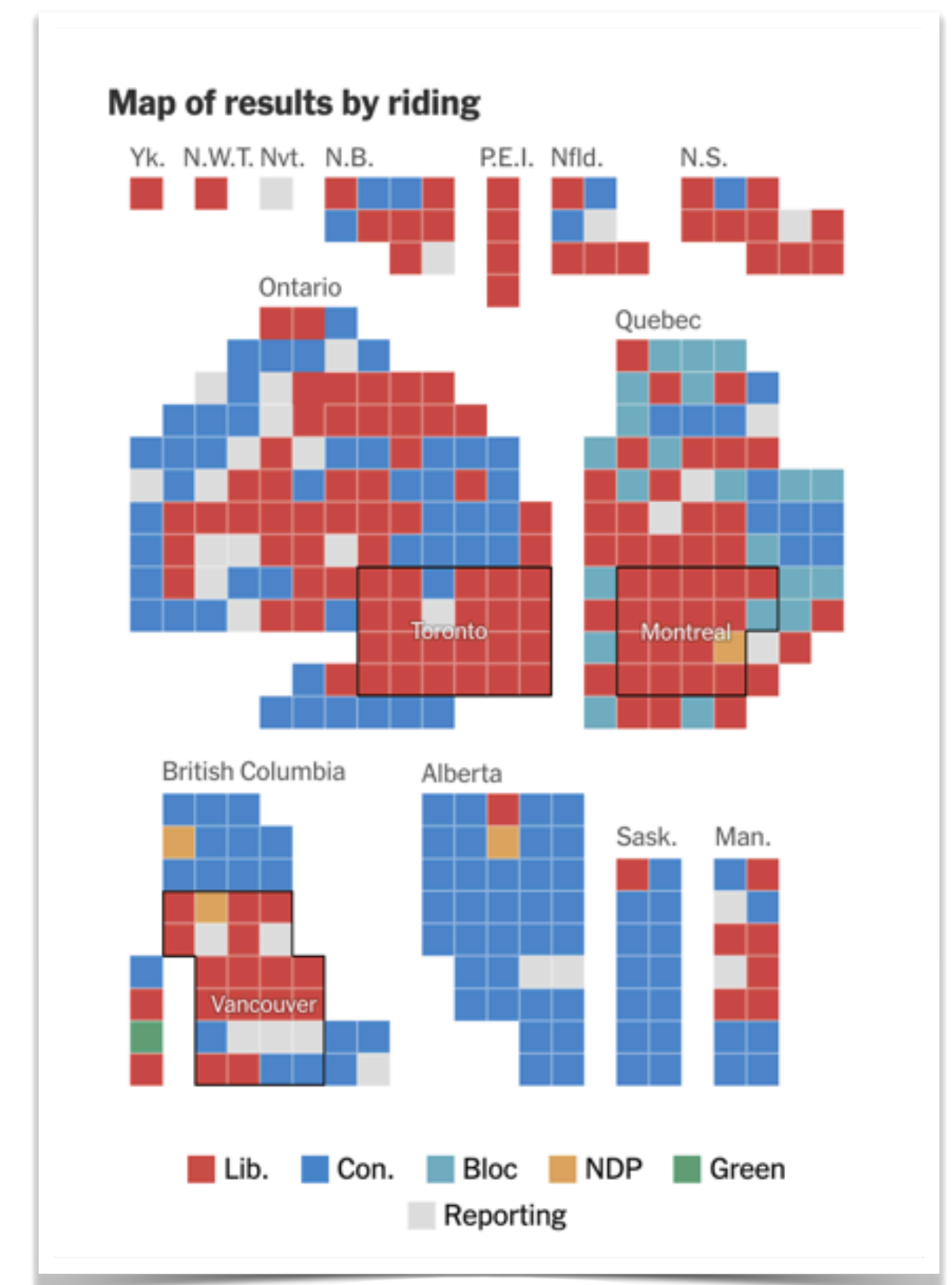
Vargas

- **Robert Vargas** is an urban sociologist here at U of C
- His book *Wounded City* examines districting and community in a neighborhood called Little Village
- Argues LV has no voice in local government because it is not only cut up *erratically*, but cut up *differently* every time, by contrast with influential neighborhoods like Bridgeport that are highly stable



Nelson

- Garrett Nelson is a historical geographer, now runs the map collection at the Boston Public Library
- Geographical places and human groups always intertwined
- “linguistic clues”: “**region**” comes from *regio*, the domain ruled by a monarch; **landscape** derives from the Germanic *Landschaft*, a unit of self-government; a **neighborhood** is both a spatial definition used by planners as well as a group of people who share the quality of neighborliness... **politics** itself derives from *polis*, referring to the Greek form of social organization within a city’s walls”
- But this is changing: “With the rise of modern forms of transportation and communication, the intensifying complexities of industrial integration, and the spread of diasporic populations, the “primary community” consisting of face-to-face relationships faded in importance.”

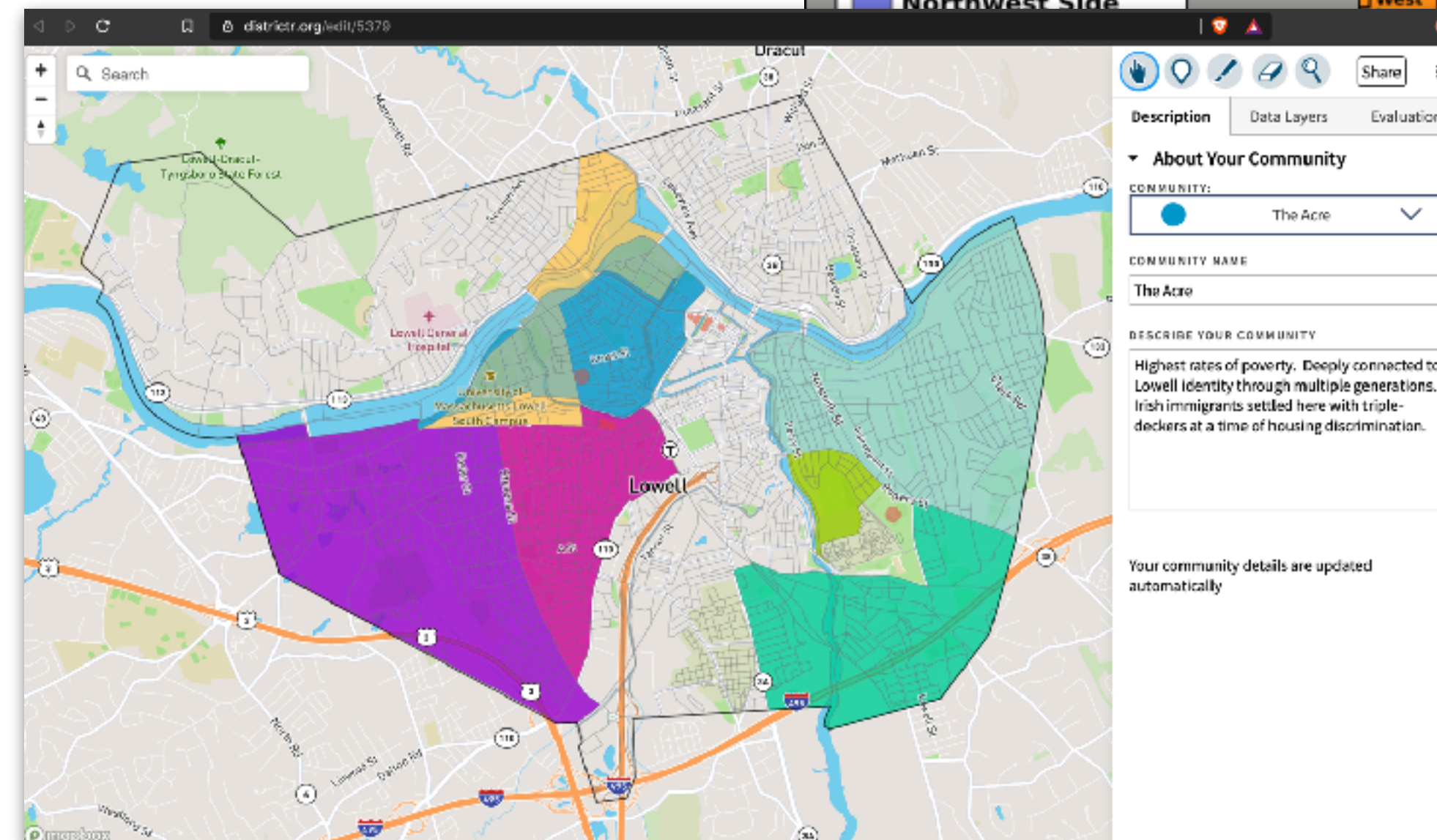


Turning community into data

- Elite/academic: Chicago's 77 community areas (pretty robust, but doesn't include Little Village!)
- Elite/governmental: Post hoc community descriptions
- Participatory: Bottom-up, crowdsourced

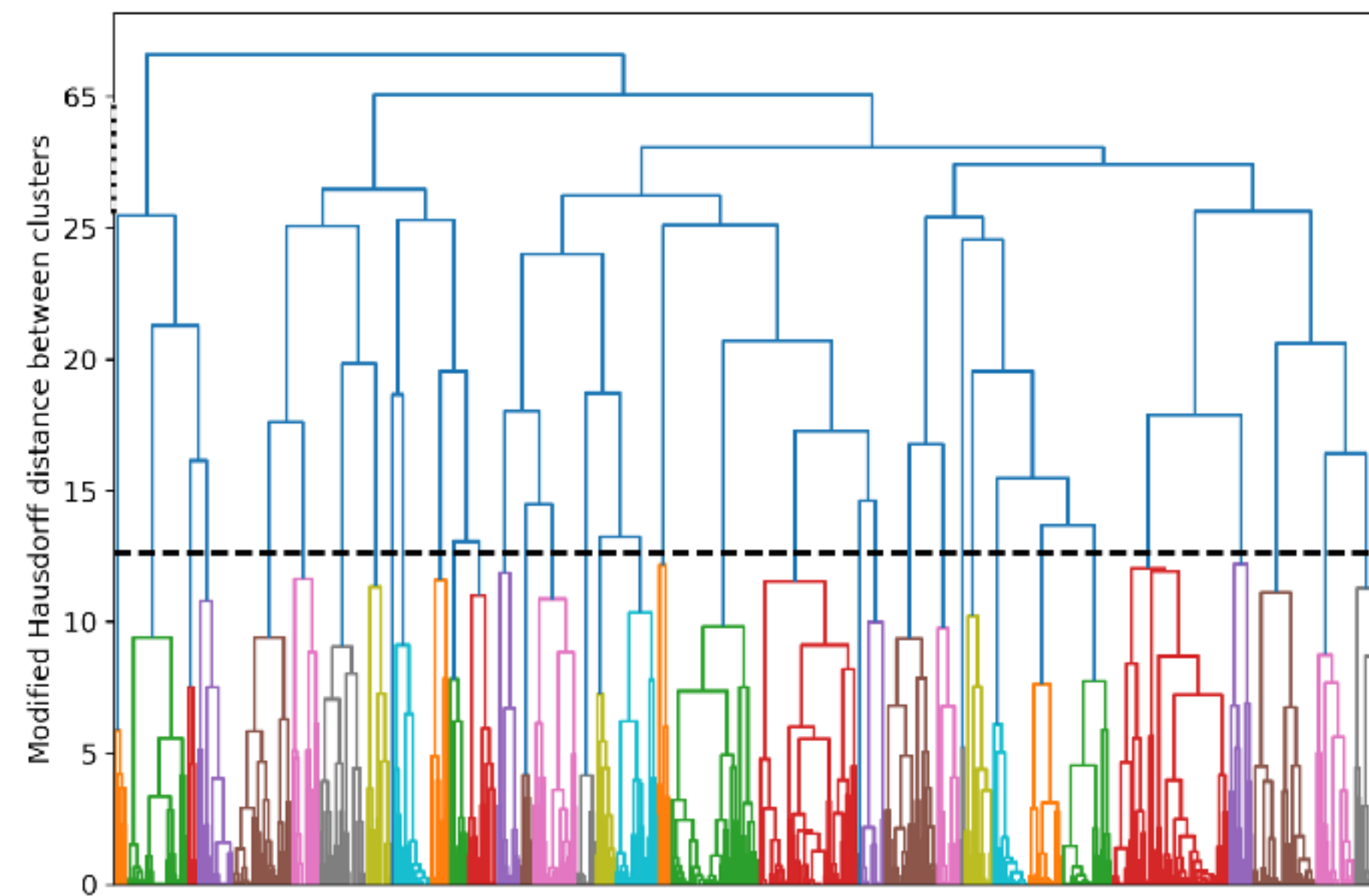


COI Theme	Count
Environmental concerns	495
Common culture/cultural community	440
Recreation	251
Fire danger/services	220
Ethnic community	164
High-tech industry	104
Aerospace industry	97
Religious community	62
Air quality	38

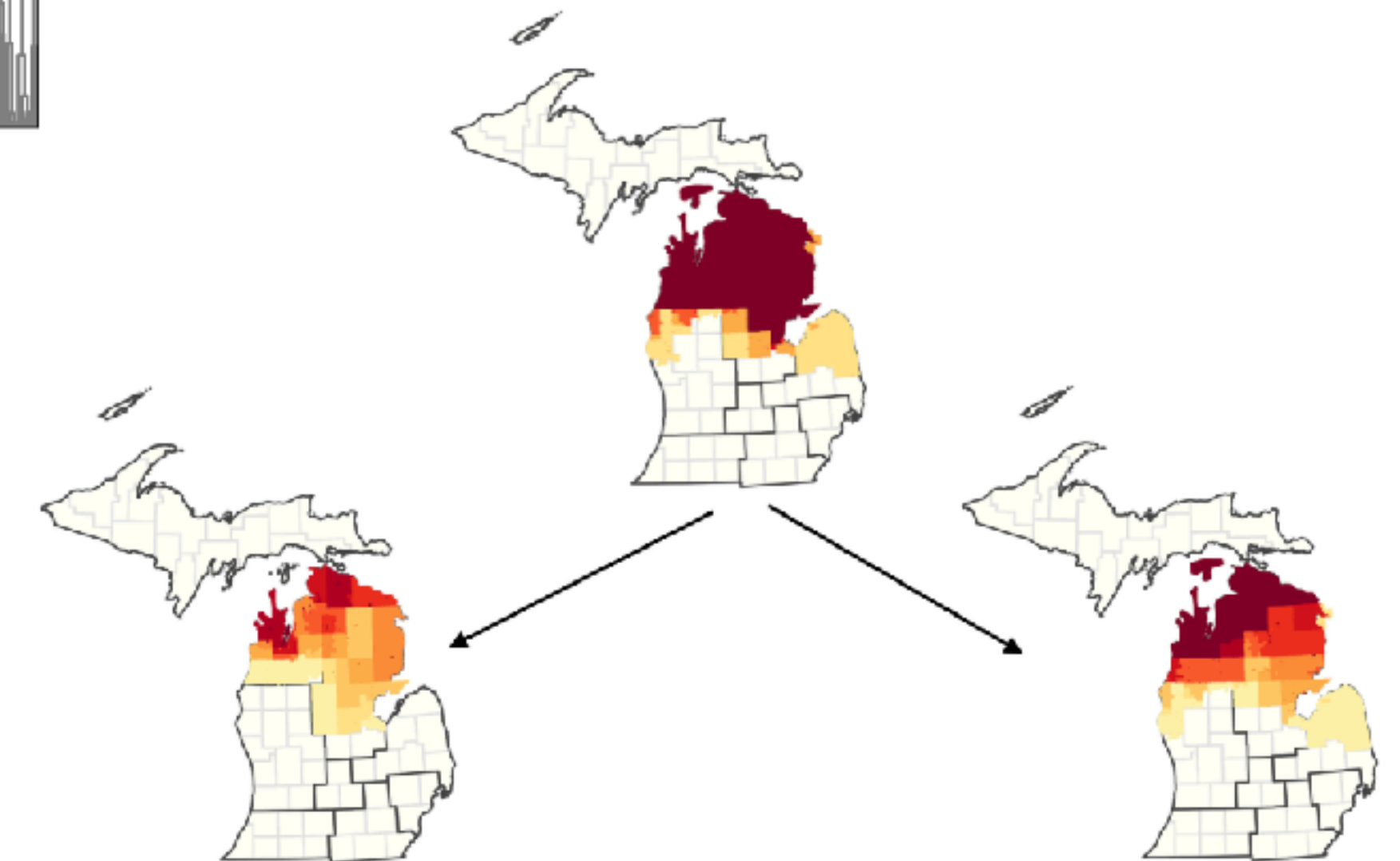


Districts, communities, and scale

- How could you measure whether a plan respects a set of communities?
- One obvious idea is splittedness (entropy!)
- But this is difficult when some are large and some are small
- My lab came up with a measurement of COI integrity:
 1. Group the communities into geographical **clusters**, preferably several dozen
 2. Check if the **language** descriptions are harmonious. If not, make sub-clusters.
 3. Sum up over the clusters a **score** from 0 to 1: 1 if the cluster is mostly whole inside a district, with a fractional reward if districts are mostly whole inside the cluster

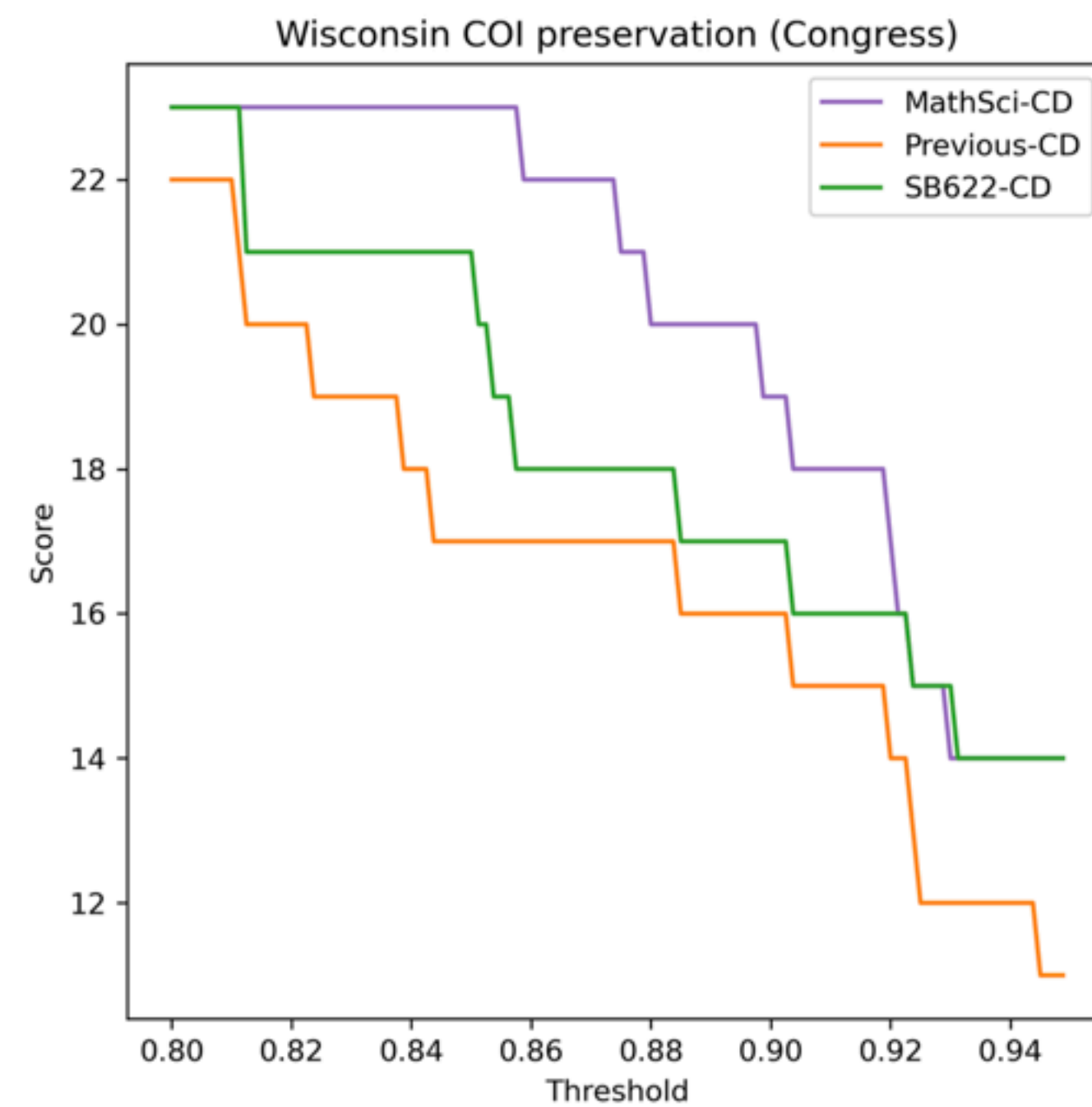


Cluster C-18, "Upper Mitten"



outdoor and
recreational spaces,
environment, and
agriculture

shared values
and identity,
including religion



what was this course
about,
in the end?

Where's Little Village? A story of place.

Close to settlement homes, late 19th c

Industry: Reaper Works (labor history!), telephone equipment, shipping, coal

Part of South Lawndale community area (1920s)

Close to 1942 public housing (Lawndale Gardens)

Integrated when Dan Ryan Expressway displaced Black and Mexican neighborhoods in 1950s

Renamed in 1960s as Lawndale got Blacker in Great Migration, new name from ethnic White residents (Czech, Polish)

74% Hispanic by 1980

Had Blue Line since 1902, lost in 2004, met with community protests, got new Pink Line in 2006

Community protest got coal plants shut down 2012

.....

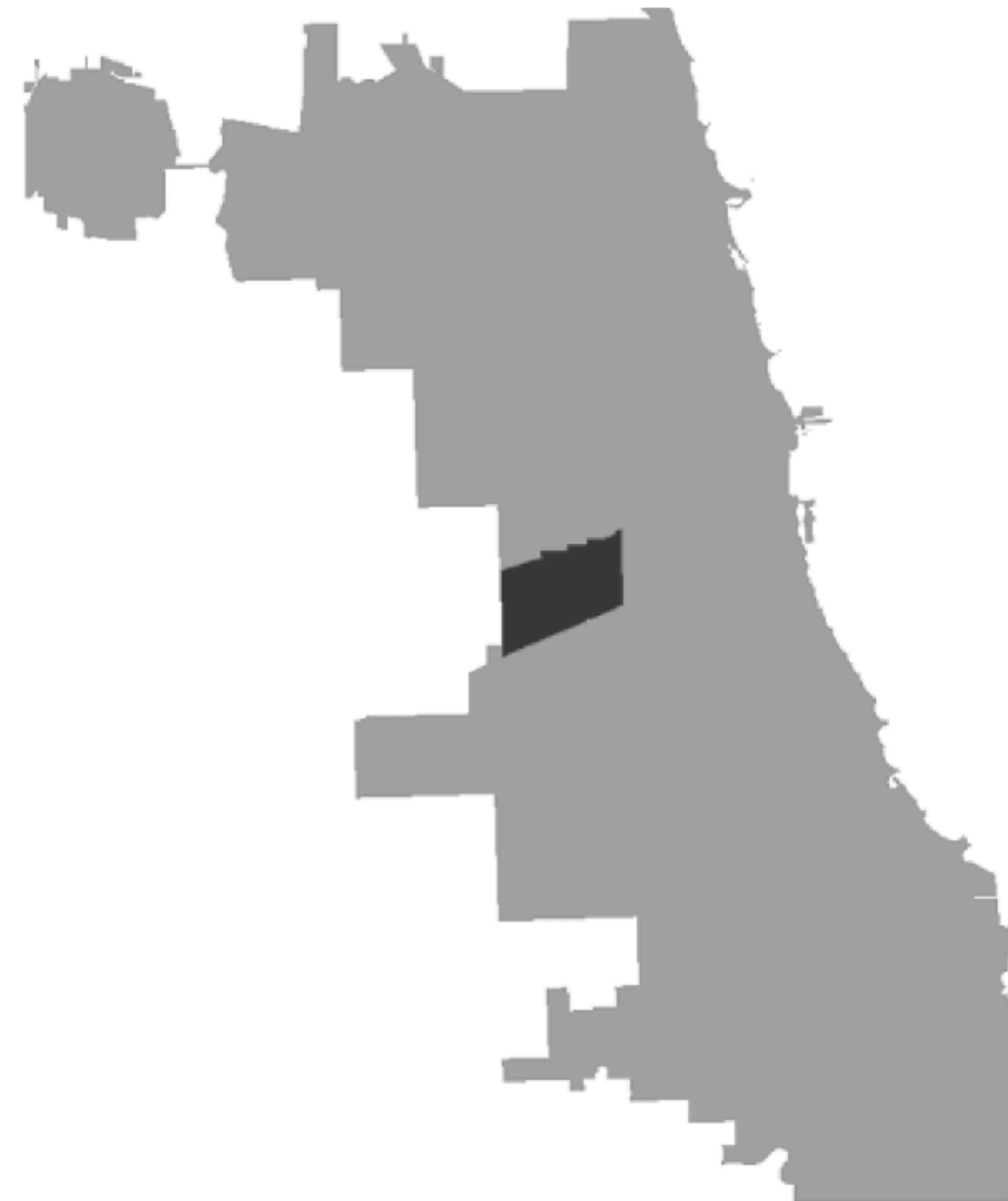
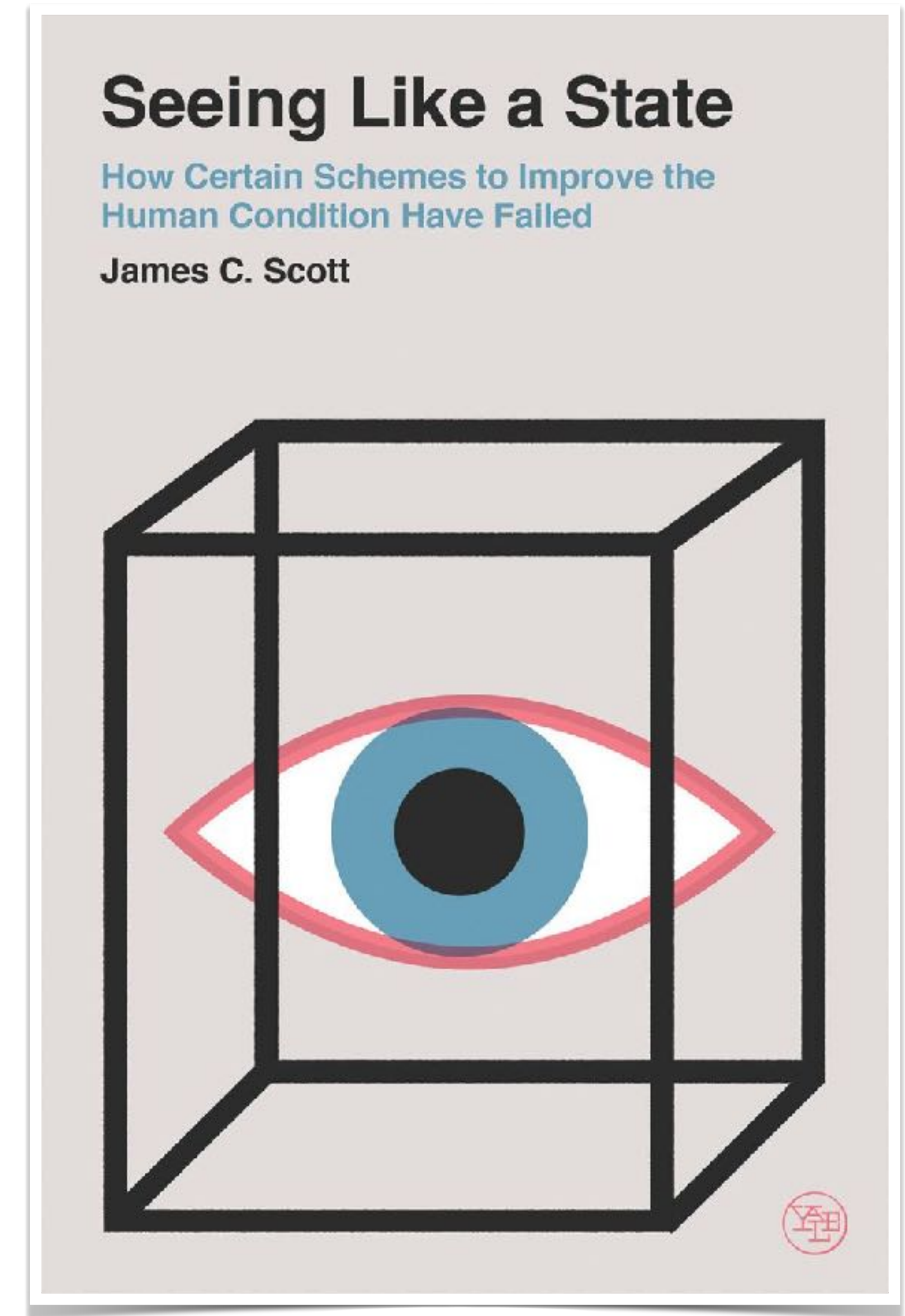


FIGURE 1.1 Location of Little Village in Chicago.

Source: Map produced by author.

Data and the State

- Control requires simplification
- Simplification can be kinds of people or descriptions of place, schemas are muscular vehicles of conceptual organization
- State power mobilizes resources and constrains actors, but it's always accompanied by “boundary work”
- We have to “operationalize” what we want to measure, and the measurement matters — there is a thirst for **thin** and **flat** description
- We've tried to look at restoring **thick** description, **critical** mechanism design, and a **qualitative** toolkit



thank you!